Why Splines ?

$$f(x) = \frac{1}{1+25x^2}$$

Table : Six equidistantly spaced points in [-1, 1]

x	$y = \frac{1}{1+25x^2}$
-1.0	0.038461
-0.6	0.1
-0.2	0.5
0.2	0.5
0.6	0.1
1.0	0.038461

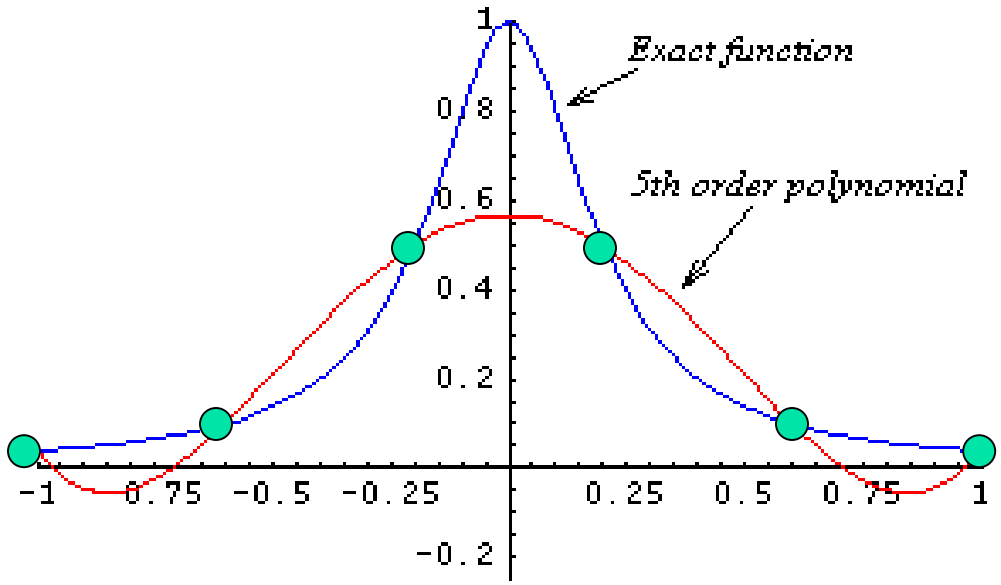


Figure : 5th order polynomial vs. exact function

Why Splines ?

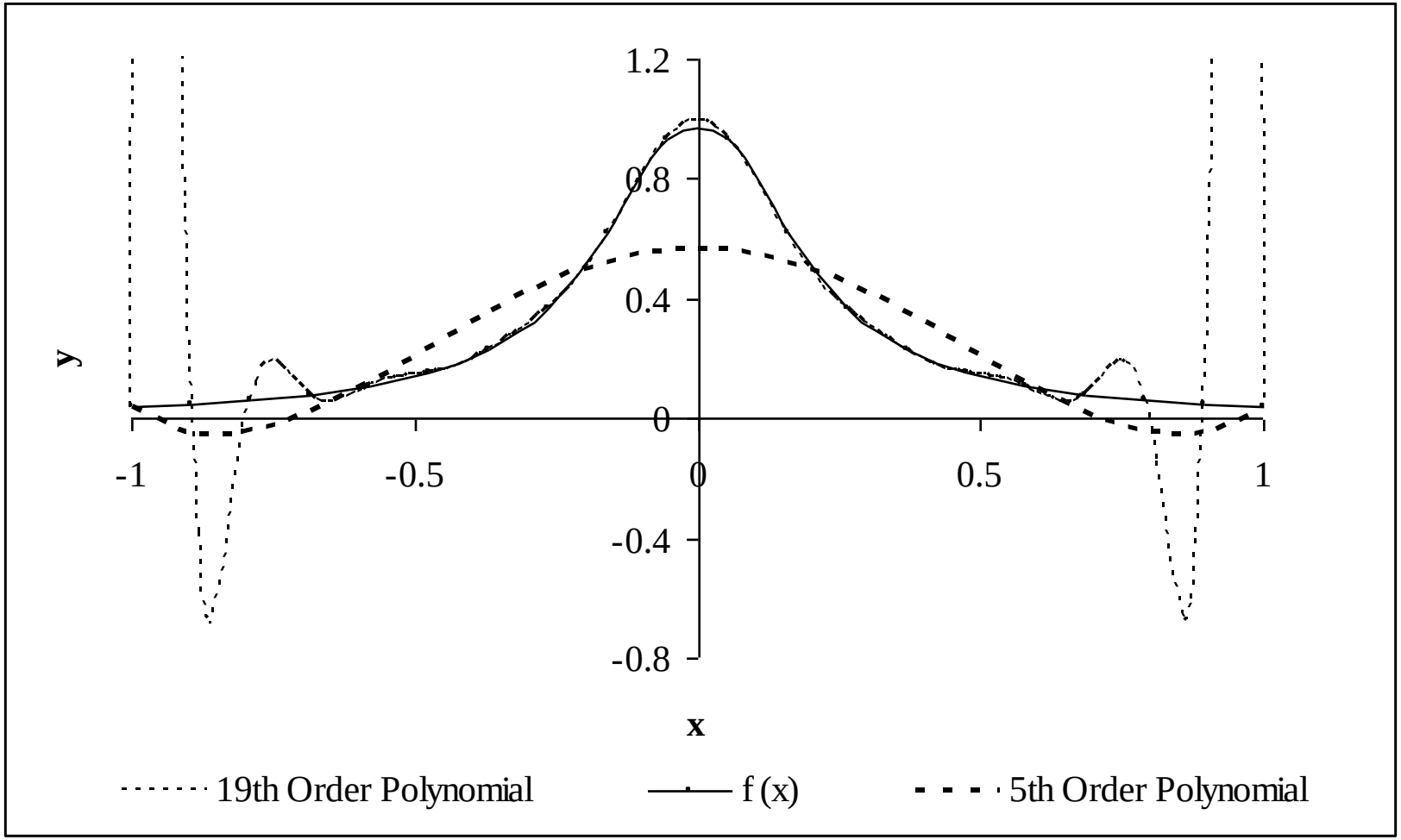


Figure : Higher order polynomial interpolation is a bad idea

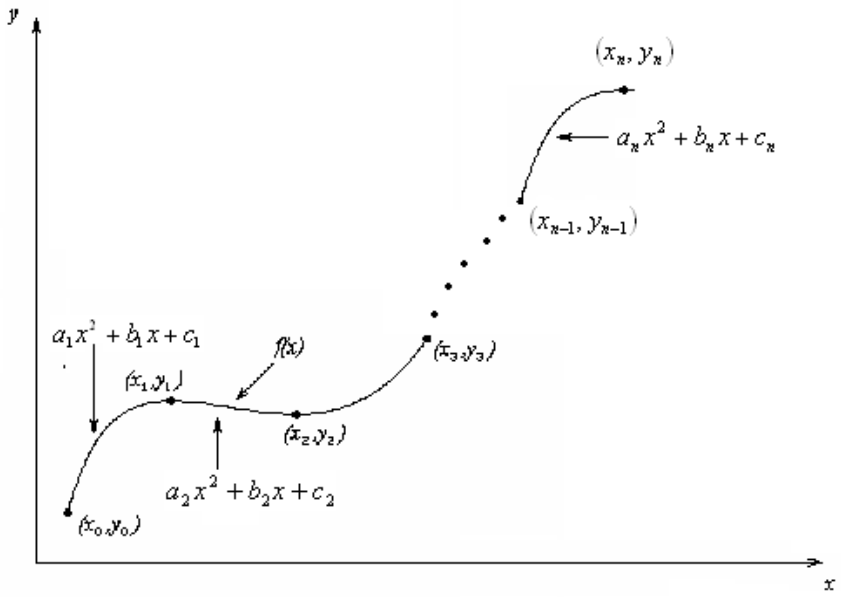
Definition of Spline

- Piecewise polynomial: To divide the entire range of points into sub-intervals. $[x_i, x_{i+1}]$, i=0,1,2, … n for n+1 points
- These polynomials may exhibits discontinuity at the interpolating points(geometric)/at the junctions(parametric).
- It is possible to construct piecewise polynomials that prevent discontinuities. (1st order or 2nd order continuity)
- This type of curve is called spline function or spline.
- It can be quadratic/cubic spline (cubic is popular)

Quadratic Spline

Given $(x_0, y_0), (x_1, y_1), \ldots\ldots, (x_{n-1}, y_{n-1}), (x_n, y_n)$, fit quadratic splines through the data. The splines are given by

$$
\begin{aligned}
f(x) &= a_1 x^2 + b_1 x + c_1, && x_0 \le x \le x_1 \\
&= a_2 x^2 + b_2 x + c_2, && x_1 \le x \le x_2 \\
&\;\;\vdots \\
&= a_n x^2 + b_n x + c_n, && x_{n-1} \le x \le x_n
\end{aligned}
$$



Find a_i, b_i, c_i, $i = 1, 2, \ldots, n$

Quadratic Spline (contd)

Each quadratic spline goes through two consecutive data points

$$a_1 x_0^2 + b_1 x_0 + c_1 = f(x_0)$$

$$a_1 x_1^2 + b_1 x_1 + c_1 = f(x_1)$$

$$\vdots$$

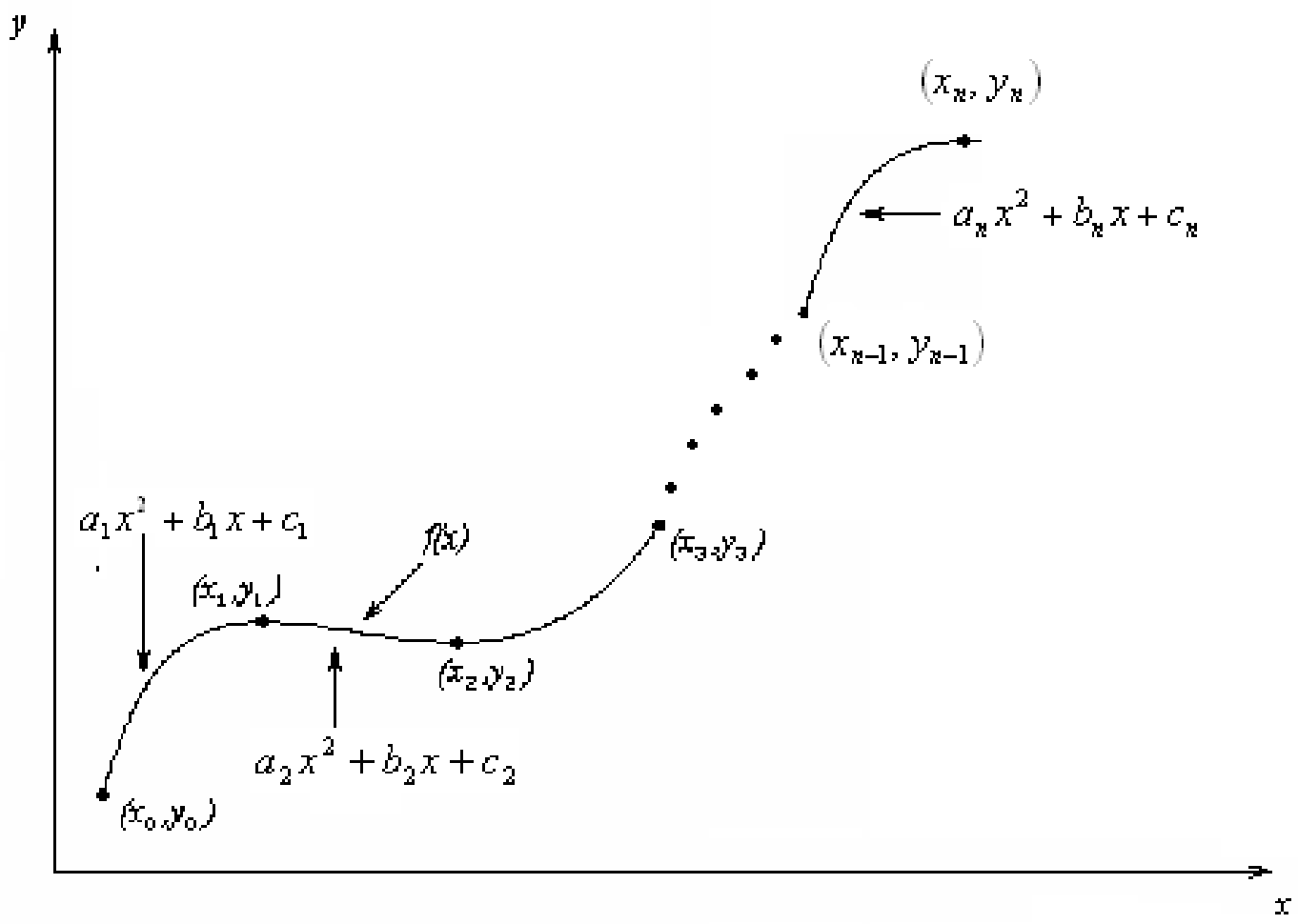
$$a_i x_{i-1}^2 + b_i x_{i-1} + c_i = f(x_{i-1})$$

$$a_i x_i^2 + b_i x_i + c_i = f(x_i)$$

$$\vdots$$

$$a_n x_{n-1}^2 + b_n x_{n-1} + c_n = f(x_{n-1})$$

$$a_n x_n^2 + b_n x_n + c_n = f(x_n)$$



This condition gives 2n equations

Quadratic Splines (contd)

The first derivatives of two quadratic splines are continuous at the interior points.

For example, the derivative of the first spline

$$a_1x^2 + b_1x + c_1 \quad \text{is} \quad 2a_1x + b_1$$

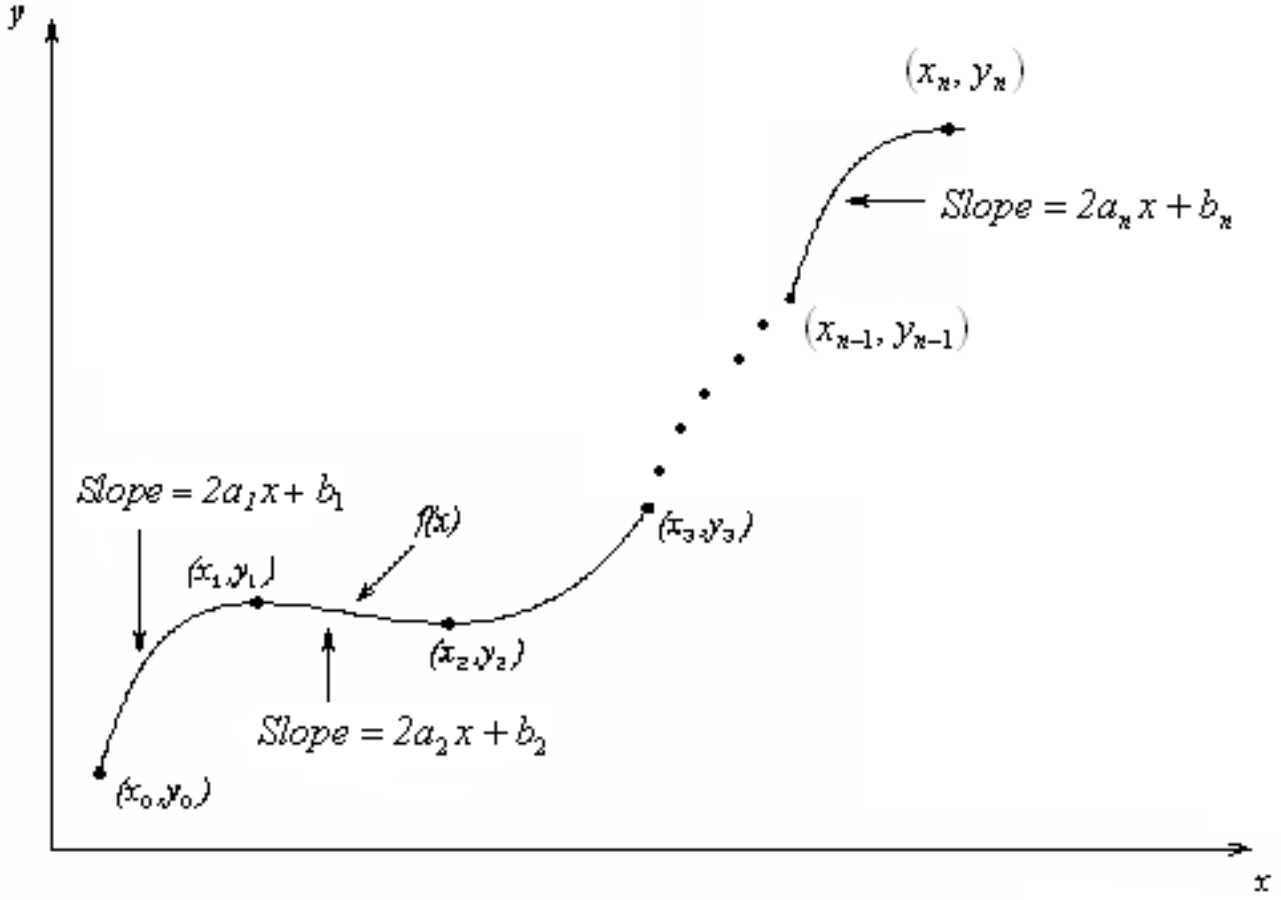
The derivative of the second spline

$$a_2x^2 + b_2x + c_2 \quad \text{is} \quad 2a_2x + b_2$$

and the two are equal at $x = x_1$ giving

$$2a_1x_1 + b_1 = 2a_2x_1 + b_2$$

$$2a_1x_1 + b_1 - 2a_2x_1 - b_2 = 0$$



Quadratic Splines (contd)

Similarly at the other interior points,

$$2a_2x_2 + b_2 - 2a_3x_2 - b_3 = 0$$

.

.

.

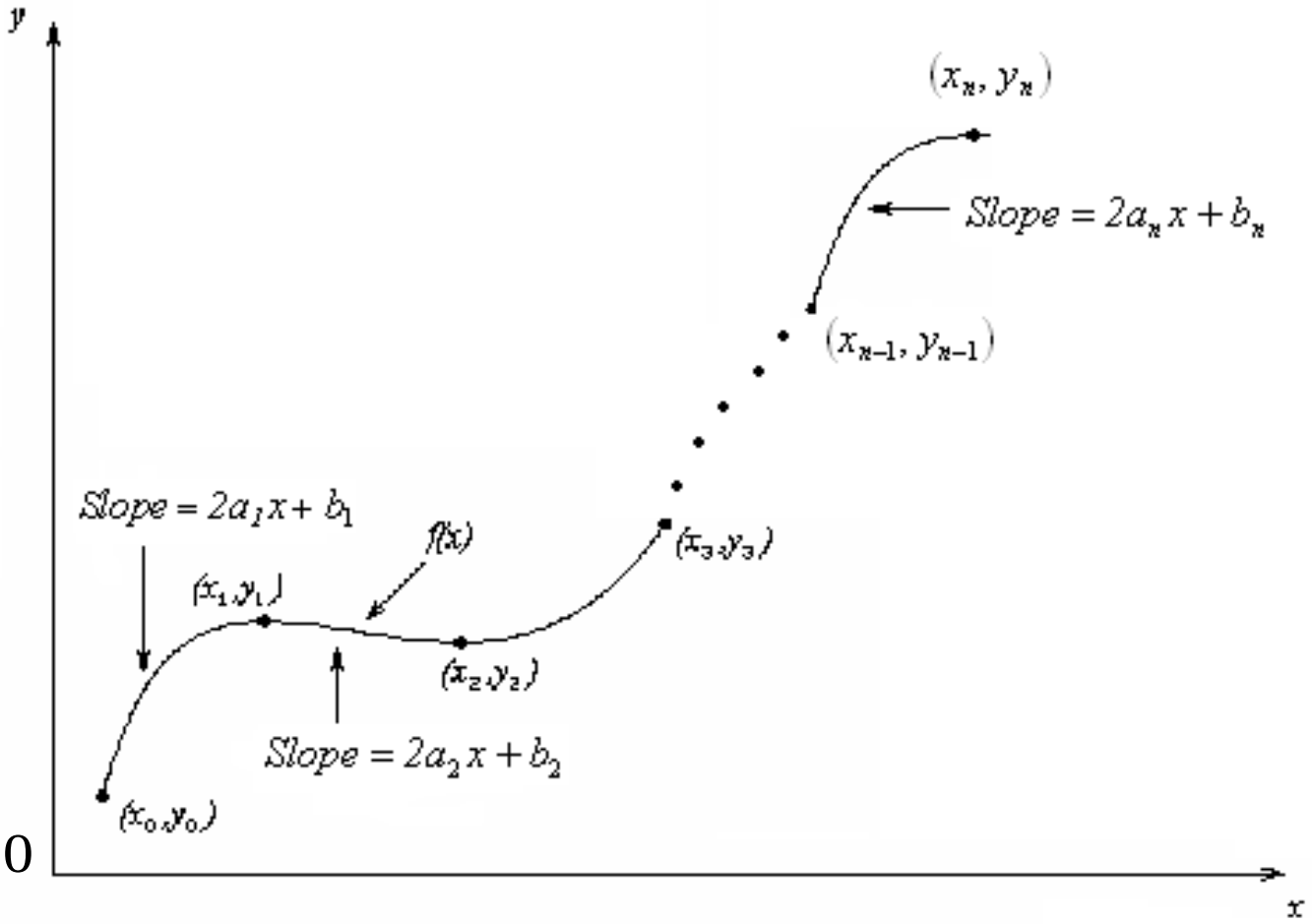
$$2a_ix_i + b_i - 2a_{i+1}x_i - b_{i+1} = 0$$

.

.

.

$$2a_{n-1}x_{n-1} + b_{n-1} - 2a_nx_{n-1} - b_n = 0$$



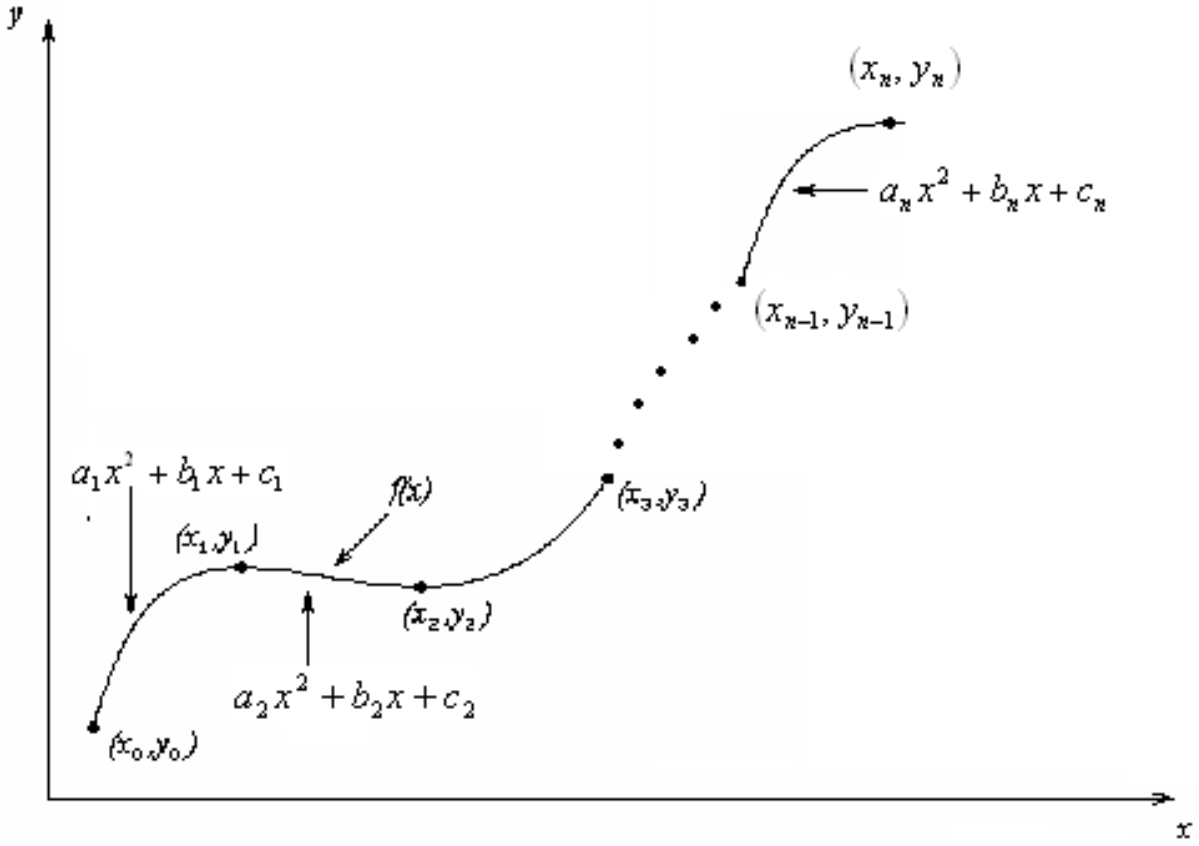
We have (n-1) such equations. The total number of equations is $(2n) + (n-1) = (3n-1)$.

We can assume that the first spline is linear, that is $a_1 = 0$

Quadratic Splines (contd)

This gives us ‘3n’ equations and ‘3n’ unknowns. Once we find the ‘3n’ constants, we can find the function at any value of ‘x’ using the splines,

$$
\begin{aligned}
f(x) &= a_1x^2 + b_1x + c_1, && x_0 \le x \le x_1 \\
&= a_2x^2 + b_2x + c_2, && x_1 \le x \le x_2 \\
&\;\;\vdots \\
&= a_nx^2 + b_nx + c_n, && x_{n-1} \le x \le x_n
\end{aligned}
$$



$$
\begin{vmatrix}
x_0^2 & x_0 & 1 & 0 & 0 & 0 & . & . & . \\
x_1^2 & x_1 & 1 & 0 & 0 & 0 & . & . & . \\
0 & 0 & 0 & x_1^2 & x_1 & 1 & . & . & . \\
. & . & . & . & . & . & . & . & . \\
. & . & . & . & . & 0 & x_n^2 & x_n & 1 \\
. & . & . & . & . & 0 & x_n^2 & x_n & 1 \\
2x_1 & 1 & 0 & -2x_1 & -1 & 0 & 0 & . & 0 \\
. & . & . & . & . & . & . & . & . \\
 & & & 2x_{n-1} & 1 & 0 & -2x_{n-1} & -1 & 0 \\
1 & 0 & 0 & 0 & & & & & 0
\end{vmatrix}
\times
\begin{vmatrix}
a_1 \\ b_1 \\ c_1 \\ . \\ . \\ . \\ . \\ a_n \\ b_n \\ c_n
\end{vmatrix}
=
\begin{vmatrix}
f(x_0) \\ f(x_1) \\ . \\ . \\ . \\ f(x_n) \\ 0 \\ . \\ 0 \\ 0
\end{vmatrix}
$$

Quadratic Spline Example

The upward velocity of a rocket is given as a function of time. Using quadratic splines

a) Find the velocity at t=16 seconds
b) Find the acceleration at t=16 seconds
c) Find the distance covered between t=11 and t=16 seconds

Table Velocity as a function of time

t (s)	$v(t)$ (m/s)
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

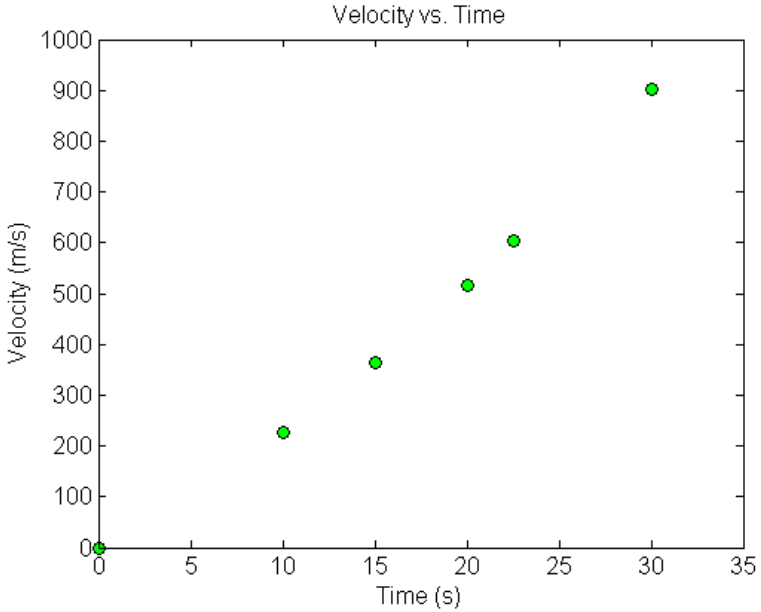


Figure. Velocity vs. time data for the rocket example

Solution

$$
\begin{aligned}
v(t) &= a_1 t^2 + b_1 t + c_1, && 0 \le t \le 10 \\
&= a_2 t^2 + b_2 t + c_2, && 10 \le t \le 15 \\
&= a_3 t^2 + b_3 t + c_3, && 15 \le t \le 20 \\
&= a_4 t^2 + b_4 t + c_4, && 20 \le t \le 22.5 \\
&= a_5 t^2 + b_5 t + c_5, && 22.5 \le t \le 30
\end{aligned}
$$

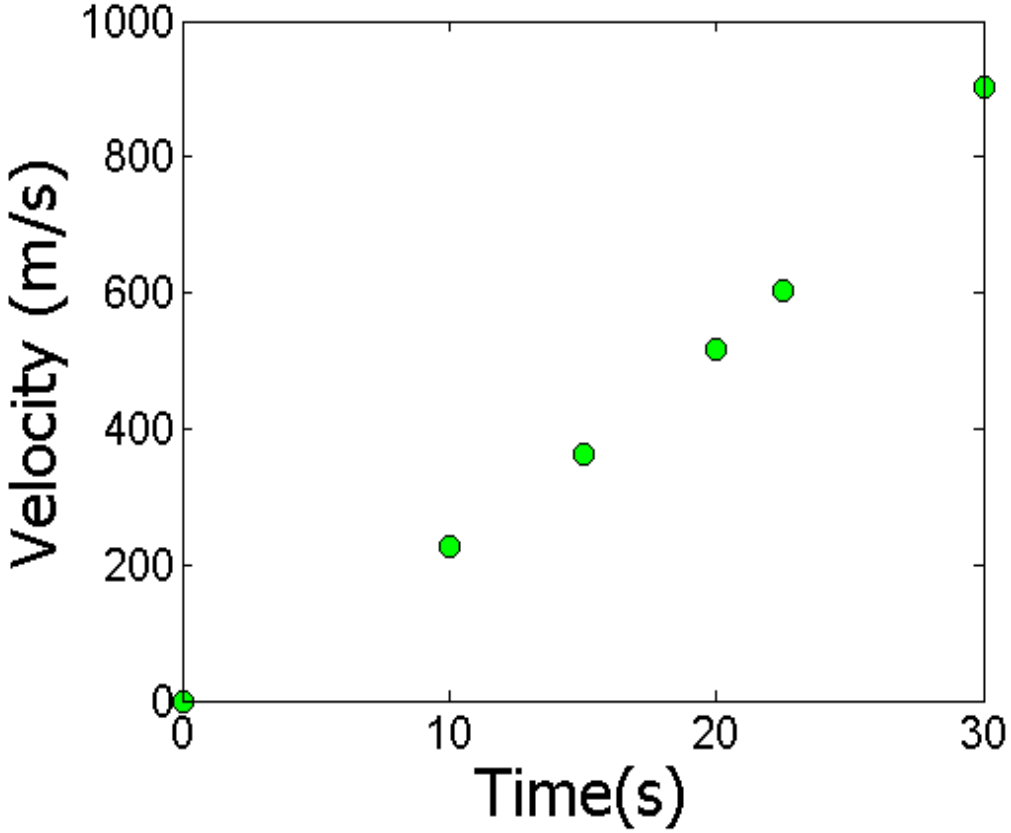
Let us set up the equations

Each Spline Goes Through Two Consecutive Data Points

$$v(t) = a_1 t^2 + b_1 t + c_1, \quad 0 \le t \le 10$$

$$a_1 (0)^2 + b_1 (0) + c_1 = 0$$

$$a_1 (10)^2 + b_1 (10) + c_1 = 227.04$$



Each Spline Goes Through Two Consecutive Data Points

t	v(t)
s	m/s
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

$$a_2(10)^2 + b_2(10) + c_2 = 227.04$$

$$a_2(15)^2 + b_2(15) + c_2 = 362.78$$

$$a_3(15)^2 + b_3(15) + c_3 = 362.78$$

$$a_3(20)^2 + b_3(20) + c_3 = 517.35$$

$$a_4(20)^2 + b_4(20) + c_4 = 517.35$$

$$a_4(22.5)^2 + b_4(22.5) + c_4 = 602.97$$

$$a_5(22.5)^2 + b_5(22.5) + c_5 = 602.97$$

$$a_5(30)^2 + b_5(30) + c_5 = 901.67$$

Derivatives are Continuous at the junction Points

$$v(t) = a_1 t^2 + b_1 t + c_1, \quad 0 \le t \le 10$$

$$= a_2 t^2 + b_2 t + c_2, 10 \le t \le 15$$

$$\left.\frac{d}{dt}\left(a_1 t^2 + b_1 t + c_1\right)\right|_{t=10} = \left.\frac{d}{dt}\left(a_2 t^2 + b_2 t + c_2\right)\right|_{t=10}$$

$$\left.\left(2a_1 t + b_1\right)\right|_{t=10} = \left.\left(2a_2 t + b_2\right)\right|_{t=10}$$

$$2a_1(10) + b_1 = 2a_2(10) + b_2$$

$$20a_1 + b_1 - 20a_2 - b_2 = 0$$

Derivatives are continuous at Interior Data Points

At t=10 $$2a_1(10)+b_1-2a_2(10)-b_2=0$$

At t=15 $$2a_2(15)+b_2-2a_3(15)-b_3=0$$

At t=20 $$2a_3(20)+b_3-2a_4(20)-b_4=0$$

At t=22.5 $$2a_4(22.5)+b_4-2a_5(22.5)-b_5=0$$

Last Equation

$$a_1 = 0$$

Final Set of Equations

$$
\begin{bmatrix}
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
100 & 10 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 100 & 10 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 225 & 15 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 225 & 15 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 400 & 20 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 400 & 20 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 506.25 & 22.5 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 506.25 & 22.5 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 900 & 30 & 1 \\
20 & 1 & 0 & -20 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 30 & 1 & 0 & -30 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 40 & 1 & 0 & -40 & -1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 45 & 1 & 0 & -45 & -1 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}
\begin{bmatrix}
a_1 \\ b_1 \\ c_1 \\ a_2 \\ b_2 \\ c_2 \\ a_3 \\ b_3 \\ c_3 \\ a_4 \\ b_4 \\ c_4 \\ a_5 \\ b_5 \\ c_5
\end{bmatrix}
=
\begin{bmatrix}
0 \\ 227.04 \\ 227.04 \\ 362.78 \\ 362.78 \\ 517.35 \\ 517.35 \\ 602.97 \\ 602.97 \\ 901.67 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0
\end{bmatrix}
$$

Coefficients of Spline

i	a_i	b_i	c_i
1	0	22.704	0
2	0.8888	4.928	88.88
3	−0.1356	35.66	−141.61
4	1.6048	−33.956	554.55
5	0.20889	28.86	−152.13

From the

1st row: $a_1 = 0$ *and* $c_1 = 0$

2nd row:

$100a_1 + 10b_1 + c_1 = 227.04$

$=> b_1 = 22.704$

Final Solution

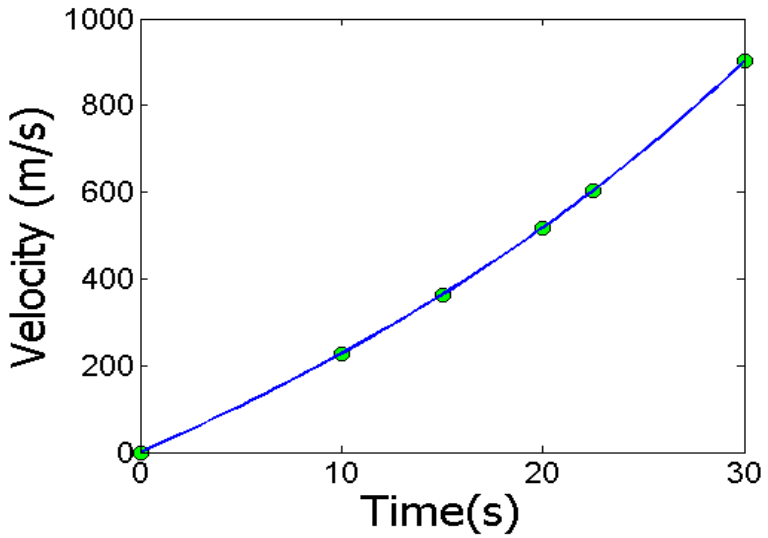
$$v(t) = 22.704t, \quad 0 \le t \le 10$$

$$= 0.8888t^2 + 4.928t + 88.88, \quad 10 \le t \le 15$$

$$= -0.1356t^2 + 35.66t - 141.61, \quad 15 \le t \le 20$$

$$= 1.6048t^2 - 33.956t + 554.55, \quad 20 \le t \le 22.5$$

$$= 0.20889t^2 + 28.86t - 152.13, \quad 22.5 \le t \le 30$$



Velocity at a Particular Point

a) Velocity at t=16

$$
\begin{aligned}
v(t) &= 22.704t, && 0 \le t \le 10 \\
&= 0.8888t^2 + 4.928t + 88.88, && 10 \le t \le 15 \\
&= -0.1356t^2 + 35.66t - 141.61, && 15 \le t \le 20 \\
&= 1.6048t^2 - 33.956t + 554.55, && 20 \le t \le 22.5 \\
&= 0.20889t^2 + 28.86t - 152.13, && 22.5 \le t \le 30
\end{aligned}
$$

$$
\begin{aligned}
v(16) &= -0.1356(16)^2 + 35.66(16) - 141.61 \\
&= 394.24 \text{ m/s}
\end{aligned}
$$

$15 \le t \le 20$